Software Requirements Specification

for

SCHOLAR PATH

Version 1.0.3 approved

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Revision History

Name	Date	Reason For Changes	Version
Abigail Diaz	11-01-2024	Added links for the documentation paperwork section 2.7	1.0.0.0
	11-01-2024	Added more detail to section 1.2 Intended Audience	1.0.0.1
Abigail Diaz	11-20-2024	External Interface Requirements- update	1.0.0.2
	11-20-2024	Pending: update use case diagram fixes	1.0.0.3

1. Introduction

Scholar Path is a navigation application for the California State University of Los Angeles main campus map. The application provides GPS navigation with real time location and classroom guided directions. The application allows users to explore student resources and amenities within the campus as well.

1.1 Purpose

Product version: 1.0.0

Product revision: 1.0.0

Product release number: 1.0.0

This document's purpose is to detail the required services the Scholar Path application aims to provide and all aspects that will be part of its user-friendly interface. Such required services are for the main purpose of enhancing how students navigate the university's campus with an interactive campus map.

1.2 Intended Audience and Reading Suggestions

Intended readers are developers, project managers, product managers, scrum masters, testers, and end-users such as clients.

The SRS contains the program's overall description followed by details of the requirements to be implemented.

For customers and end-users that only wish to understand what the program provides, it is recommended to read:

Section 1: Introduction [ALL]

Section 2: Overall Description

- 2.1 System Analysis
- 2.2 Product perspective
- 2.3 Product Functions
- 2.4 user Classes and Characteristics

For developers, product managers, scrum masters and testers that wish to understand in more detail what will be implemented, it is recommended to read the entire document with emphasis on:

Section 3: External Interface Requirements [ALL]

Section 4: Requirements Specification [ALL]

Section 5: Other Nonfunctional Requirements [ALL]

1.3 Product Scope (Abigail Diaz)

Software Products in production: Scholar Path.

The software will provide:

- Campus Navigation: Scholar Path will offer a GPS supported map with turn-by-turn directions to classrooms, study group meetings and amenities such as bathrooms, libraries, food courts and other campus locations.
- Information on amenities: Users can locate amenities such as cafes, bathrooms, libraries, bookstores and be provided with a description of the services provided, hours of operations, contact information and accessibility information.
- Personalized Routes: Based on the user's favored classes, Scholar Path will provide directed routes to classrooms upon their selection.
- Group meetings: Students can locate and navigate to group meetings that they are a part of.
- Account creation: Students can create their own accounts to have access to the maps, along with access to their personalized classroom selections.
- Guest Login: Users may access the program as a guest by proving an email address, name and agreeing to the terms of service.

Scholar Path will not provide the following:

- Will not provide academic services such as course registration or transportation services.
- Scholar path will not provide 3D modeling of the inside of buildings.
- Scholar Path will not provide inside mapping of the buildings.
- Scholar Path will not route the location of classrooms with a directed line inside buildings, instead the classroom location will be the closest approximate from outside the building.

Usage of the software once released:

A navigation tool to find classes, locate and be informed about campus resources and find group meetings with fellow students.

Objectives of the software:

Scholar Path is a campus navigator designed to assist students, faculty, and visitors to efficiently find resources and locations within campus. The application will provide real-time navigation, building information, amenities, and resources to enhance the means students and staff navigate the university's campus.

Benefits of the software:

- Enhanced efficiency in getting to classes or meetings, easier access to campus resources and amenities.
- Simplified navigation of the campus due to reduced time spent searching for offices, group meetings, classrooms, or facilities.
- A more user-friendly navigation of campus, improving students and staff experience when exploring the campus and reducing confusion during the first week of school.

Goal of the Software: To improve campus accessibility by facilitating the quick and efficient way students find locations and resources.

1.4 Definitions, Acronyms, and Abbreviations (LG)

GPS: Global Positioning System

API: Application Programming Interface

UI: User Interface

Map Integration: Combining map data with real-time GPS functionality for navigation

CIN: Campus Identification Number

1.5 References

- List of all documents referenced in the SRS: SDD
- List of any Web addresses referenced in the SRS: N/A

2. Overall Description

Scholarpath is designed to assist college students, staff, and visitors in navigating the campus efficiently. The program will provide an interactive map that displays the locations of classrooms, buildings and amenities (e.g., cafeterias, food courts, libraries). It will also offer features such as GPS-based navigation, study group location, guest access to the application and account creation.

2.1 System Analysis (Joshua Hanscom)

This product identifies the issue of classes, buildings, and amenities being difficult to find and navigate for students new to CSULA. To solve this problem, the product is designed to guide students to their desired destination via a website containing an intuitive/digestible interface that allows the user to respond to the information provided, ultimately leading them to their destination and solving the original problem.

Major Technical Hurdles

1. Extracting useful information including classroom numbers, professor names, and building codes from public CSULA data sheets.
2. Mapping coordinate or Geohash positions to points of interest on campus
    a. Without an already provided list of coordinate positions of buildings, classrooms, and other amenities, the development team must, to some extent, manually record the positions of each point of interest to be displayed on the map interface.
3. Constructing a relational Database Structure
    a. Each table within must preserve an appropriate relationship to the other tables in the database that can be reflected when referenced in JavaScript via classes.
    b. Tables will use a minimum of 2NF formalization.
4. Creating Object-Oriented JavaScript to reflect the database's structure
    a. To effectively pull necessary information from the database and provide functionality to the user, the JavaScript used for the logic design of the webpage should be structured in a way that allows simple integration with the database.
5. Communicating DB Data between frontend and backend via servers.

Associated Solutions

1. Formatting the available spreadsheets to exclude fields unnecessary to the product and sending these to our own spreadsheet to later be imported into an SQL database. So far, we have pulled the necessary information into our own spreadsheet.
2. Have team members manually record the latitude and longitude of each point of interest via a mobile device. This may result in having a smaller part of the campus available in the map interface by product delivery date.
3. A database structure has been developed to allow relations between tables when necessary. Please refer to section 8. This was done by considering the necessary fields for our product and the manner in which they can interact with object-oriented logic code.
4. Classes that interact with the database and each other should reflect the different tables within the database. This allows for the categorization of information and its use in functionality to be more streamlined when moving from backend to front-end.
5. Developing an API Server w/ DB OO Classes designed for frontend/ backend compatibility

2.2 Product Perspective (Joshua Hanscom)

Scholarpath is a standalone product designed to enhance campus navigation by utilizing technologies such as APIs and databases. The program relies on Leaflet API to create an interactive map and the Navigator API for required geolocation services. These APIs provide reliable and efficient performance, as well as compatibility with modern browsers. The program will also rely on a MySQL database to store the user profile data and classroom location and information.

Comparison to Similar Systems

There are several existing map applications, such as Google Maps and Waze, that also offer navigation and information on location. However, these systems do not provide more detailed, campus-specific information such as classroom numbers, more precise location of classrooms or group meeting coordinating services. Even though the other apps offer similar navigation services, this remains more focused on the general user, while Scholarpath is designed with college students and staff in mind as the main user demographic.

Context Diagram

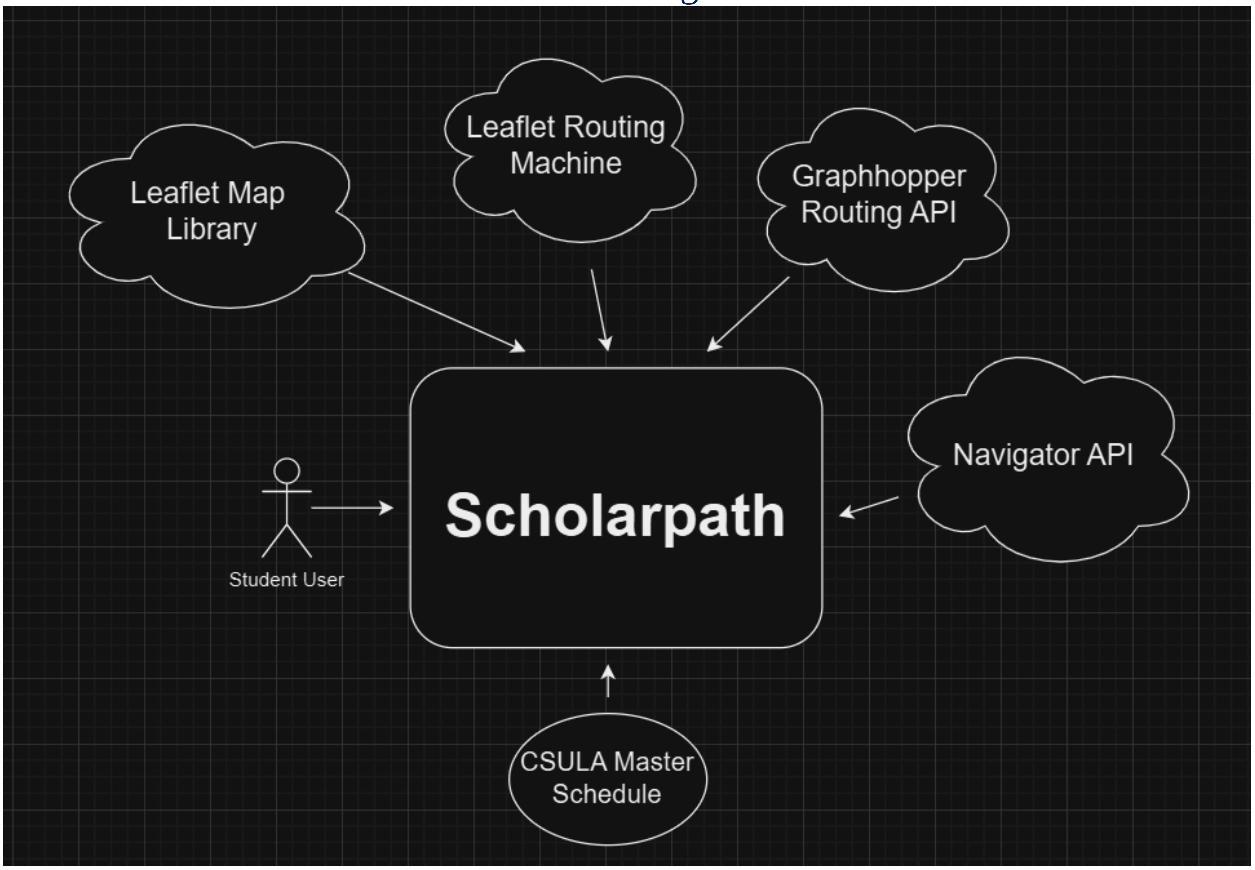
The Context Diagram below illustrates the relationships between Scholarpath and its external systems.

User Devices: Students, staff, and visitors access the application via web browsers.

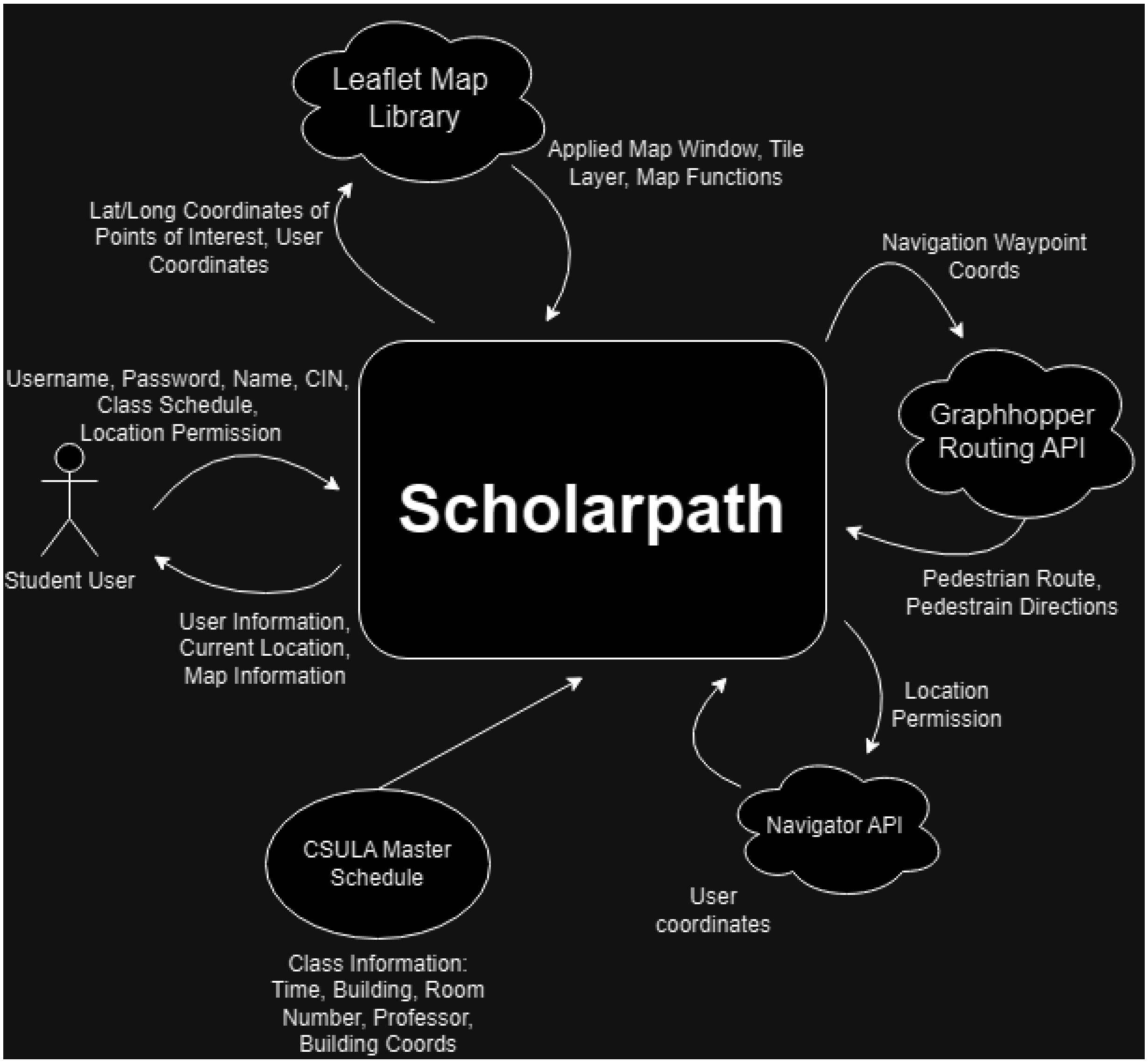
Campus Data Systems: Real-time data is retrieved from systems like facilities management and scheduling from the CSULA Master Schedule.

External APIs: Utilizes Leaflet API for map rendering and Navigator API for geolocation.

Context Diagram



The student will interact directly with the product. Our product, including front-end and back-end functionality, will interact with the external integrations and data to ultimately deliver a website that solves the original issue in a digestible way to the user.



Data Flow Diagram

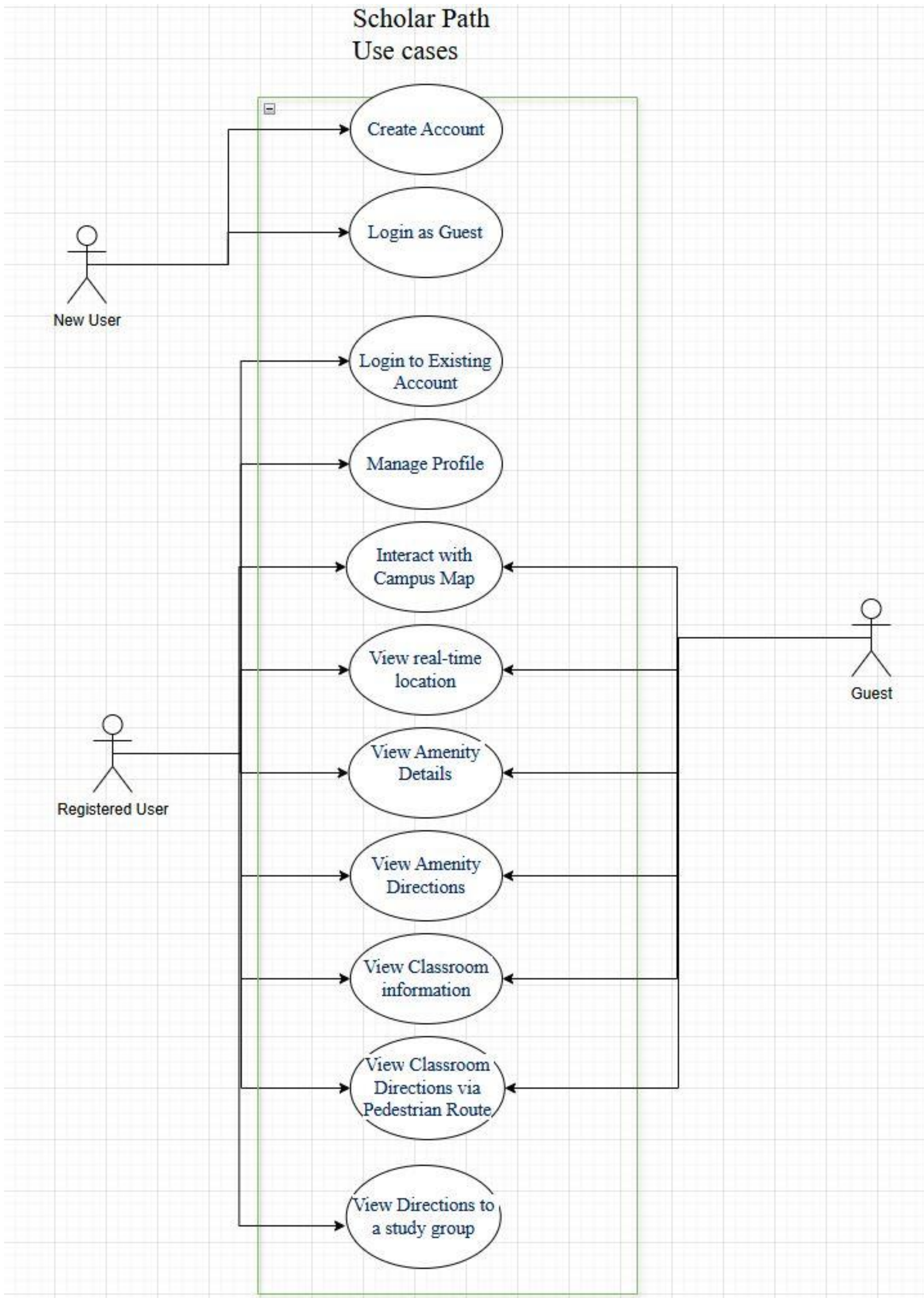
The product will interact with each external source, where no external sources are communicating with each other. The user will provide necessary information, the product will then check its database created from the information of the CSULA Master Schedule. Next, the product will show the allowed location to the Navigator API, with Navigator providing the user's coordinates in return. With these coordinates and those of the points of interest within the product, they are displayed by the leaflet map library by assigning the data to the appropriate centering and marker coordinates. These coordinates are also sent to the Graphopper API in conjunction with Leaflet Routing Machine (LRM) to overlay a pedestrian route and turn-by-turn directions on the map interface. The information within the map and profile interfaces are then returned to the user.

2.3 Product Functions

Primary actors: Students/ Faculty with an account and Students/ Faculty without an account

Use Cases

1. Create Account: The user can create an account to access more features in the application. (Kenia)
2. Login as Guest: Users who have not created an account can access limited features. (Leonardo Granados)
3. Login to Existing Account: Users can login into their account. (Kenia)
4. See Location on Map: Users can view their real time location in relation to the campus map by seeing the location marker on the map. (Joshua Hanscom)
5. View Amenity Directions: Users can search for amenity locations and obtain directions. (Ahin Huq)
6. View Amenity Details: Users can view amenity information. (Abigail Diaz)
7. Select Classroom Destination: Users can select from a list of their classes to redirect pedestrian routing to the coordinates of the selected destination (Joshua Hanscom).
8. Search buildings and reroute directions to destination (Joshua Hanscom)
9. View Classroom Directions via Pedestrian Route: Users can search for their classroom location. (Joshua Hanscom)
10. View Classroom information: Users can see details of their classes. (Jonathan Little)
11. View Directions to a study group: User can locate a study group they belong to. (Charlie Kaing)


Scholar Path
Use cases
Create Account
Login as Guest
New User
Login to Existing Account
Manage Profile
Interact with Campus Map
View real-time location
Guest
View Amenity Details
Registered User
View Amenity Directions
View Classroom information
View Classroom Directions via Pedestrian Route
View Directions to a study group

2.4 User Classes and Characteristics (LG)

- Students (New/Transfer): Primary users will be using the app to navigate between classes/amenities and make it easier to transition into their new environment.
- Students (Recurring): Secondary users will use the app to navigate between classes and buildings they haven’t been in.
- Campus Visitors: May use the app to navigate through campus/events.
- Faculty/Staff: Potential users for navigating to meetings or other campus locations

2.5 Operating Environment (LG)

- Mobile Web Application: Runs on Android and iOS browsers.
- Desktop Web Application: Compatible with modern web browsers like Chrome, Firefox, Edge, etc.

The Software's main environment will consist of all of Cal State LA campus ground.

2.6 Design and Implementation Constraints (LG)

1. The application depends on real-time data from GPS and the course database.
2. The system must ensure data security, allowing only authorized users (students and staff) to access campus maps and course information.
3. Testing on different hardware could result in different bugs found to be worked on.

2.7 User Documentation (LG)

Installation of server: Will be in the program’s GitHub, PDF Format

Database connection: Will be in the program’s GitHub, PDF Format

2.8 Assumptions and Dependencies (LG)

1. The software assumes that users will have a stable internet connection.
2. GPS accuracy depends on the hardware of the user's device.
3. The software assumes that the user has hardware to run the application.
4. The course database should be updated regularly by the university administration.
5. The course database should be updated correctly by the administration.
6. Faculty should update the room immediately if a room change occurs.

7. Faculty should update the room if class will be on zoom.

2.9 Apportioning of Requirements

Requirements that might be delayed until future versions of the system:

Sharing routes or real-time location with friends or groups.

3. External Interface Requirements

The Scholar Path web application is designed to provide a comprehensive campus navigation and resource tool for students on campus. To achieve this goal the application will incorporate external interfaces to support user interactions through web browsers, GPS and geolocation services, mapping APIs, server, and database communications.

3.1 User Interfaces

Primary Interface: The Scholar Path application is designed primarily as a web application that runs on smartphones and desktop computers.

Input Methods

The primary input methods will include:

- Touchscreen: Users interact with the application by tapping on buttons, icons, and the campus map.
  - Touchscreen Compatibility: If accessed on a tablet or touchscreen laptop, the functionality won't be compromised.
- Keyboard on smartphones: On-screen keyboards will be available when necessary (search feature).
- Physical keyboard on desktop
  - Mouse and Keyboard: The web interface will use mouse and keyboard inputs.
- Browser-Based Interface: Web-based interface for desktops, laptops and smartphones.

3.2 Hardware Interfaces

The Scholar Path application is primarily web-based. Its implementation is based on hardware components found in desktop and mobile devices. This section outlines the relationship between the hardware interfaces and the application for proper implementation.

1. Client Device (Desktop, Laptop, Tablet, Smartphone)

Supported Device Types

Any device with a web browser, including desktops, laptops, tablets, and smartphones.

The program interacts with the user-end device's input/output capabilities through the web browser:

Needed for user interface and handling user input:

- HTML
- CSS
- JavaScript

2. GPS Navigation (Mobile Devices)

Supported Device Types: Mobile smartphones with built-in GPS hardware.

The web application accesses the device's GPS through an API, enabling the app to obtain the user's real-time location for navigation purposes.

3.3 Software Interfaces

The Scholar Path web application will run with different interfaces to achieve its functional requirements. Here we outline the required software interfaces, their versions, sources, and a description of the APIs used for data exchange.

1. Web Server

- Name: Apache HTTP Server
- Version: 2.4.62
- Source: Apache Software Foundation

Communication:

The Scholar Path application will communicate with the Apache server using the HTTP/HTTPS protocol. To allow requests from the user's browser to reach the server for data retrieval and manipulation.

2. Database Management System

- Name: MySQL → DB to Frontend API (Joshua Hanscom) / SQL to OO JS model
- Version: 8.0.36
- Source: Oracle Corporation

Communication:

The application will use MySQL database using SQL queries to store and retrieve data.

(Joshua Hanscom Begin)

3. Server Runtime Software
   - Name: Node.js
   - Version: v20.16.0
   - Source: Node.js
4. Map Routing Software
   - Name: Leaflet Routing Machine (LRM)
   - Version: 3.2.12
   - Source: https://www.liedman.net/leaflet-routing-machine/
5. Pedestrian Routing API
   - Name: Graphhopper API
   - Version: 10.0
   - Source: Graphopper

(Joshua Hanscom End)

3.4 Communications Interfaces

The Scholar Path web program incorporates necessary communications interfaces to implement data exchanges between users and servers. This section outlines the requirements related to communication interfaces.

1. Web Browser Communications

Protocol: HTTP/HTTPS

Description:

The application will communicate over HTTP/HTTPS protocols to securely transmit data between the client and the server. This is accomplished with HTTPS user information security protections during data exchanges.

Message Formatting:

Data exchanged will be formatted in JSON (JavaScript Object Notation) for API responses and requests.

Security: in progress, possibly TLS (Transport Layer Security)

2. Database Interfaces

The application will use MySQL database to store location data and route history.

4. Requirements Specification

This section contains all the necessary software requirements with enough detail to allow designers to accurately design the software to satisfy those requirements, and to allow testers of the software to verify that all requirements have been satisfied. The requirements include a description of every input to the system, every output, and all functions performed by the system in response to an input or output.

1. **Functional Requirements**
   Given that the user wants to use our services, *when* the user is on our login page, *then*:

1.1. The user shall be able to create an account on the login page. (Kenia and Leonard)

1.1.1. The user should be able to register with an email and password of their choosing.

1.1.2. The user shall be notified of the account creation success.

1.1.3. The user should be able to login with the email and password used during the account creation.

1.2. The system shall allow the user to have guest access. (Leonard)

1.2.1. The system shall have a guest access page besides the login page.

1.2.2. The access page shall have an email text field.

1.2.3. The guest access page shall have the terms and services.

1.2.4. The guest access page shall give access to the map once the user logs in as guest.

1.3. The user shall be able to login on the login page. (Kenia and Leonard)

1.3.1. The login page should be neatly presented in a legible way.

1.3.2. The login page takes a username and password for map access.

1.3.3. If the password or username is wrong, the user shall be notified.

1.3.4. The user has the option to change their password.

1.3.5. The map should be visible once the user logs in.

1.4. The users should be able to manage their profile settings on the landing page. after logging in)

1.4.1. Users should be able to update their profile under the profile tab, which gives access to profile settings.

1.4.1.1. Users should be able to change their passwords once in profile settings.

1.4.2. The user should be able to view favorite classes under the "my favorites" tab.

1.4.3. Users shall find it easy to navigate the landing page.

1.5. The user shall be able to interact with the campus map.

1.5.1. The system shall respond to user actions (e.g., clicking icons, zooming) within 2 seconds under normal load conditions.

1.5.2. The system should provide a user-friendly interface that adheres to established design principles.

1.5.3. The system may include a help section or tutorial for first-time users to facilitate understanding of the features.

1.6. The user should be able to view their real time location in relation to the campus map. (Joshua Hanscom).

1.6.1. *Given* that the user has logged in and has allowed access to location, *when* the user is on the map page, *then*:

1.6.1.1. The application shall allow the user to see their current location via a marker on the map so that they know their location relative to the campus.

1.6.1.2. The application shall provide accurate positioning of the location marker such that the user can understand the placement of the location marker on the map.

1.6.1.3. The application shall update the position of the location marker in relation to the change in the user's position such that the user is able to see their change in position reflected on the map interface.

1.6.2. The system shall display the layout of the campus accurately, reflecting the real locations of each building.
1.6.3. The system shall allow users to scroll up, down, and sideways across the map without restrictions.
1.6.4. The system shall allow users to zoom in and out on the map while maintaining a consistent display of buildings and icons.
1.6.5. The system should provide a smooth transition effect when zooming or panning to enhance user experience.

1.7. The system shall allow users to view amenity information.

1.7.1. The system shall provide clickable amenity icons across the map that display the amenity title and a simple description in a text box when the icon is clicked.
1.7.2. The system shall ensure that the text box is adjusted based on the map zooming.
1.7.3. The system shall use black, legible text in the text box that remains clearly readable.
1.7.4. The system shall display appropriate information within the text box, including hours of operation, contact information (if any), and a photo of the amenity.
1.7.5. The text box should have a little cross on the top right so the user may close the amenity selection.

1.8. The system shall allow the user to view amenity directions. (Ahin)

1.8.1. The system shall display amenity icons on the side of the relevant buildings on the campus map.
1.8.2. The system shall provide a pop-up display titled "Amenities" that lists distinct amenity icons on the right side of the screen, but in the middle.
1.8.3. The amenity menu shall not interfere with the user's ability to interact and view the map in a significant way.
1.8.4. The system shall display the amenity title on the right next to the icons.
1.8.5. When one of the icons is clicked, it will display a list of amenities on the side, with a small photo, title and description.
1.8.6. While the list is open the user may click on one of the amenities and a button shall appear next to the amenity description in the list that says, "Go," where Amenity is the name of the selected Amenity.

1.8.6.1. The user shall have the option to close the menu by either clicking on another icon or pressing the closing x on the right top corner of the list.
1.8.6.2. If the user clicks the Go button, a route will appear to the selected amenity.
1.8.6.3. The system shall hide the list once the “Go” button is pressed for better map visibility.
1.8.6.4. A text box will appear and display “Routing to Amenity”, where Amenity is the name of the selected Amenity. A photo may appear underneath the text as well.
1.8.6.5. The user can follow the route to reach their destination.
1.8.6.6. The user may exit the navigation by clinking “Exit Navigation” on the top center of the map.
1.8.6.7. The system may include an option to filter amenities based on user preferences (e.g., food, restrooms).
1.8.6.8. The system may display all available pathways in blue if there are multiple amenities in the same building, allowing users to select one.
1.8.6.8.1. The system may keep unselected pathways visible in gray to indicate they are still available for selection.

1.9. The system shall allow users to view their class information. (Jon little and Joshua Hanscom)

1.9.1. The user should be able to view and interact with their classroom data via any relevant interface.
1.9.2. The database shall hold the class information for every user and store the locations between sessions.
1.9.3. The database shall have a list of classes offered this semester linked to the class, so that a User can look up their classes and add it to their class list.

1.10. The system shall allow the user to view classroom directions (Joshua Hanscom).

1.10.1. Given that the user has logged in and has allowed access to location, when the user is on the map page, then:
1.10.2. The application shall provide a list of the user’s class titles in a selection list.
1.10.3. The application shall allow where, upon clicking on one of the list elements, that element will appear active in the selection list.
1.10.4. The application shall allow where, upon clicking on one of the list elements, the destination on the map will change to that of the element.
1.10.5. The application shall allow where, upon clicking on one of the list elements, the path will reroute in response to the new destination.

1.11. The system shall allow the user to view study group locations.

1.11.1. The system shall provide a search bar to find specific study groups by keyword.
1.11.2. The system shall display the user’s GPS location to show the closest study group locations.
1.11.3. The system shall provide an appropriate user interface design for the group list option.
1.11.4. The system shall have a selection to view the location of the study group on the map.
1.11.5. The system shall allow the user to see directions to where the selected study group is located.
1.11.6. The system may notify the user about upcoming study groups that they are a part of.

4.1 Functional Requirements

Functional requirements describe functions the Scholar Path system must provide to achieve its purpose. These requirements prescribe the functions that a system must provide and define the processes and operations used to link inputs to outcomes.

4.1.1 User Account Creation and Login

Description:
The system shall allow users to create accounts, log in, and access campus map features or use guest access with limited functionality.

4.1.1.1 Account Creation

4.1.1.1.1 The system shall allow the user to create an account on the login page.

Who: Students, University Staff or users who want personalized services the system provides.

Why: To have secure access to the program’s services and sensitive data and manage user-specific settings or data within a more controlled environment.

4.1.1.1.2 The system shall allow the user to register with an email and password of their choosing.

Who: Users who do not have an account under the email address provided.

Why: To allow flexible account creation and to implement cybersecurity measures.

4.1.1.1.3 The system shall notify the user of the account creation success.

Who: New users who provided the new email address for username purposes and password.

Why: To confirm successful registration and avoid user confusion about whether the registration was successful or not.

4.1.1.2 Registered User Login

4.1.1.2.1 The system shall allow users to log in using the email address and password provided during the account creation.

Who: Registered users.

Why: To authenticate users and grant access to their data and personal profile.

4.1.1.2.2 The system shall notify users if the provided username or password is incorrect.

Who: Registered users who input incorrect email address or password.

Why: To improve usability by offering immediate feedback of errors.

4.1.1.2.3 The system shall allow users to change their password via the profile settings.

Who: Registered users who are logged in to the system.

Why: To ensure security, since only the authentic user may change their own password and to provide the user control over their account details.

4.1.1.3 Guest Access

4.1.1.3.1 The system shall allow guest users to access the campus map without creating an account.

Who: Temporary users who do not wish to register.

Why: To provide flexibility and accommodate casual usage.

4.1.1.3.2 The system shall have a guest access page besides the login page.

Who: Potential guest users.

Why: To avoid confusion by clearly separating guest access page from user login and registration.

4.1.1.3.3 The guest access page shall have an email text field and display the terms and services.

Who: Guest users.

Why: To ensure guest users are accounted for based on email address provided and to facilitate limited functionality access.

4.1.1.3.4 The system shall allow guest users to access the map upon agreeing to terms and providing an email.

Who: Guest users.

Why: To grant basic access while maintaining a degree of accountability.

4.1.2 Campus Map Interaction

Description:
The system shall allow users to interact with the campus map for navigation, real-time location, and accessing amenity information.

4.1.2.1 Real-time user location

4.1.2.1.1 The system shall display the user's current location on the map as a marker.

Who: Logged-in users who enable location tracking.

Why: To help users orient themselves on campus.

4.1.2.1.2 The system shall update the location marker in real-time as the user moves.

Who: Walking users navigating the campus.

Why: To ensure the map reflects the user's position accurately.

4.1.2.1.3 The system shall accurately position the marker relative to campus buildings and pathways.

Who: Users with GPS.

Why: To clearly allow the user to view their location on campus and avoid confusion.

4.1.2.2 Map Navigation

4.1.2.2.1 The system shall allow users to scroll up, down, and sideways across the map.

Who: All users (guest, registered users)

Why: To provide full access to the campus layout.

4.1.2.2.2 The system shall allow users to zoom in and out on the map while maintaining consistent display quality.

Who: All users.

Why: To maintain visibility and visualization of specific areas.

4.1.2.2.3 The system should provide smooth transitions when zooming or panning.

Who: All users.

Why: To create a seamless user experience.

4.1.2.3 Amenity Information

4.1.2.3.1 The system shall provide clickable amenity icons across the map.

Who: Users looking for campus amenities.

Why: To allow quick access to amenity details.

4.1.2.3.2 The system shall display amenity details (title, description, hours, contact info) in a pop-up box when an icon is clicked.

Who: Users interacting with amenities on the map.

Why: To provide the amenity information in a user-friendly format.

4.1.2.3.3 The pop-up box shall have a close button in the top-right corner.

Who: All users.

Why: To allow users to exit details easily.

4.1.3 Landing Page Features

Description:
After login, the system shall provide personalized features on the landing page, including profile management and favorite locations.

4.1.3.1 Profile Management

4.1.3.1.1 The system shall allow users to manage their profile settings from the profile tab.

Who: Logged-in users.

Why: To allow personalization and updates to account details.

The system shall allow users to change their password in profile settings.

Who: Logged-in users.

Why: To ensure security.

4.1.3.2 Favorite Classes

4.1.3.1.2 The system shall allow users to view their favorite classes under the "My Favorites" tab.

Who: Logged-in users.

Why: To provide easy access to frequently visited locations.

4.1.3.1.3 The system shall allow users to update or remove favorite classes.

Who: Logged-in users.

Why: To reflect changes in user preferences.

4.1.4 System Performance and Usability

4.1.4.1 The system shall respond to user actions (e.g., clicking, zooming) within 2 seconds under normal load conditions.

Who: All users.

Why: To maintain performance expectations and avoid frustration.

4.1.4.2 The system shall provide a user-friendly interface according to standard design principles.

Who: All users.

Why: To ensure ease of use and accessibility.

4.1.4.3 The system may include a help section or tutorial for first-time users.

Who: New users unfamiliar with the system.

Why: To improve quick understanding of the system.

4.1.5 Amenity Directions

Description:
The system shall allow users to locate and navigate amenities on the campus map.

4.1.5.1 Amenity Visualization

4.1.5.1.1 The system shall display amenity icons inside of relevant buildings on the campus map.

Who: All users.

Why: To clearly indicate where amenities are located.

4.1.5.1.2 The system shall provide a pop-up display titled "Amenities" that lists distinct amenity icons on the right side of the screen, positioned in the center.

Who: Users looking for a list of available amenities.

Why: To allow users to quickly find amenities.

4.1.5.1.3 The amenity menu shall not interfere significantly with the user's ability to interact and view the map.

Who: All users.

Why: To maintain map usability while viewing amenities.

4.1.5.2 Amenity Details and Navigation

4.1.5.2.1 The system shall display the amenity title next to the respective icons in the list.

Who: All users.

Why: To provide description about each amenity.

4.1.5.2.2 When an amenity icon is clicked, the system shall display a list of amenities with a small photo, title, and description.

Who: Users selecting a specific amenity type on the map.

Why: To offer detailed information about the selected amenity.

4.1.5.2.3 The list shall include a "Go" button next to the amenity title description, starting navigation to the selected amenity.

Who: Users who wish to navigate to a specific amenity.

Why: To facilitate routing to the amenity destination.

4.1.5.2.4 The system shall hide the list once the "Go" button is clicked for better map visibility.

Who: Users navigating to amenities.

Why: To reduce distractions during navigation to the amenity.

4.1.5.2.5 The system shall display a message saying "Routing to [Amenity]" with a photo of the amenity beneath the text.

Who: All users.

Why: To confirm to the user that the navigation to the selected amenity is in progress.

4.1.5.2.6 The system shall allow the user to exit the navigation by clicking “Exit Navigation” at the top center of the map.

Who: Users who no longer wish to follow the route.

Why: To give users control/flexibility over the pace of the navigation.

4.1.5.2.7 The system may include an option to filter amenities based on user preferences (like food or restrooms).

Who: Users with specific amenities needs in mind.

Why: To allow users to find amenities faster by narrowing down options.

4.1.5.2.8 If multiple amenities are in the same building, the system may display available pathways in blue, keeping unselected pathways visible in gray.

Who: Users selecting an amenity.

Why: To indicate all available options.

4.1.6 Classroom Directions

Description:
The system shall enable users to view their class information and receive directions to their classrooms.

4.1.6.1 Classroom Data Management

4.1.6.1.1 The database shall store class information for every user and persist this data between sessions.

Who: System developers.

Why: To ensure reliability of user data.

4.1.6.1.2 The database shall include a list of all classes offered this semester, allowing users to look up and add classes to their list.

Who: Users managing their schedules.

Why: To simplify the process of personalizing class schedules.

4.1.6.2 Classroom Navigation

4.1.6.2.1 The system shall display a list of the user’s class titles as a selection list.

Who: Logged-in users.

Why: To provide quick access to class-specific information.

4.1.6.2.2 When a user clicks a class from the list, the system shall highlight the selected class in the list.

Who: Logged-in users.

Why: To indicate active selection.

4.1.6.2.3 Upon selecting a class, the system shall adjust the map to show the destination corresponding to the class location.

Who: Logged-in users.

Why: To provide a positional reference for the selected class in the campus map.

4.1.6.2.4 The system shall reroute the navigation path based on the newly selected class.

Who: Logged-in users.

Why: To adapt the route dynamically based on user selection of classes.

4.1.7 Study Group Locations

Description:
The system shall allow users to view and navigate to study group locations on campus.

4.1.7.1 Study Group Search and Visualization

4.1.7.1.1 The system shall provide a search bar to find specific study groups by keyword.

Who: Users searching for particular groups.

Why: To simplify the process of locating study groups.

4.1.7.1.2 The system shall display the user's GPS location alongside nearby study group locations.

Who: Users looking for the closest study group.

Why: To enable user to find study group more easily.

4.1.7.1.3 The system shall allow users to view the location of a study group on the map.

Who: Users selecting a specific study group.

Why: To help locate the group visually.

4.1.7.2 Study Group Navigation

4.1.7.2.1 The system shall provide directions to the selected study group location.

Who: Users navigating to a study group.

Why: To facilitate navigation to the group location.

4.1.7.2.2 The system may notify users about upcoming study groups they are part of.

Who: Users with scheduled study groups.

Why: To remind the user of the upcoming event.

4.2 Requirement for the external interface

In this article, a complete list of inputs to be processed into the software system and output, will detail that section. External interfaces include:

User interfaces: System will support mobile and desktop based interfaces. It is possible to make use of interaction consisting of touch or mouse input, and over a browser, open up maps along with other sources.

Hardware Interfaces — the application will utilize the device GPS to access location-based service Smartphones, tablets and desktops are supported devices.

Software Interface: The system Scholar Path will interface with the databases of MySQL and GPS info by Navigator API.

Other API will communicate over HTTP/HTTPS and use JSON as a data format for the payloads (Request & Response).

Mapping and Navigation Libraries: Leaflet API will be used to render interactive maps on both mobile and desktop platforms. It enables panning, zooming, and displaying map layers. GraphHopper API will extend on the user friendliness of the system by providing route calculation for navigation between user locations and destinations based on pedestrian mode.

4.3 Logical Database Specifications (Joshua Hanscom)

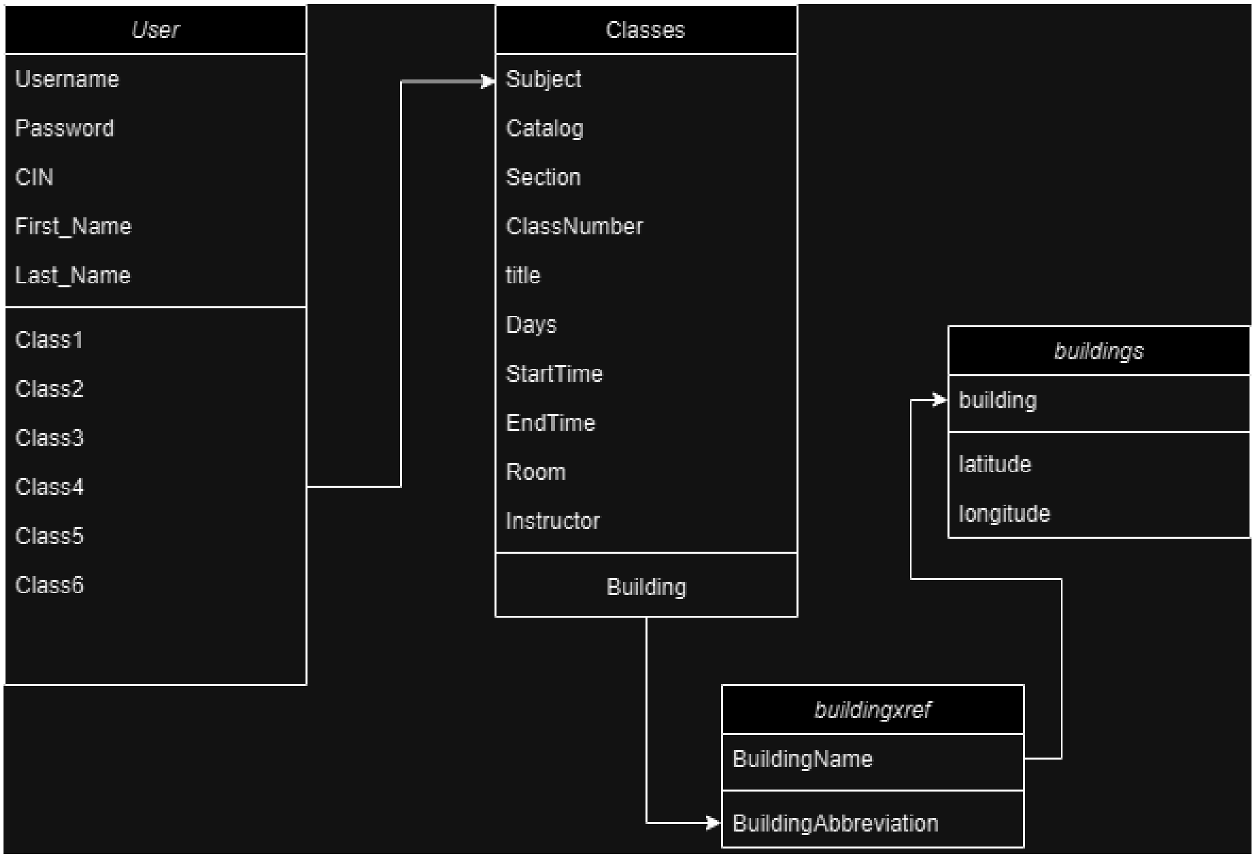
This is where the logical design of the Scholar Path database and how different entities relate to one another will be specified. Key tables include:

User Table: This table contains the user authentication, class schedule and properties.

Classes Table — Data regarding the individual class like name, schedule and related building.

Buildings Table: Associate each building with latitude and longitude→

Buildingsxref: Connects Classes and Buildings Tables



User Table

- Authenticates user, holds all class and children tables
- Accessed by map page, login page, profile page
- Must hold non-NULL values for all its fields EXCEPT classes

Classes

- Must hold non-NULL values for all its fields
- Accessed to display name and date to user via map page interface

Buildings

- Must hold non-NULL values for all its fields
- Accessed to display point of interface via map page interface

Buildingxref

- Must hold non-NULL values for all its fields
- Provides cross referencing between Classes and Buildings

4.4 Design Constraints (KS)

Specify design constraints that can be imposed by other standards, hardware limitations, etc. This should be a more technical description of the overview given in section 2.5.

- App should be capable of compatibility for different screens sizes, resolutions such as phones, iPad, computers etc.
- Memory should be optimized efficiently around 8-12 GB minimum and 16+ GB average using it efficiently to prevent excessive consumption of data.
- We will need enough RAM to handle memory-intensive tasks without crashing.
- Our app should use the minimum battery life for a device such as reducing backround processes and efficient algorithms.
- Wi-Fi connection may interfere with the app usage and the app should handle seamless user experience regarding stable connection.

5. Other Nonfunctional Requirements (Charlie Kaing)

5.1 Performance Requirements(CK)

Scholar Path uses Web browsers for maximum compatibility and ease of use for students at Cal State LA. The performance criteria are the following:

- Only one map of the campus is displayed
- Scholar Path as a web app should be able to support multiple students browsing the app. This includes being able to retrieve information for multiple students using the app. The current test criteria are around 3 simultaneous students.
- The class location on a student's schedule shown on the map can be retrieved at around 1 second of request.
- GPS should be responsive to student location as they are walking to class with no more than 1-3 seconds of latency.

5.2 Safety Requirements(CK)

Scholar Path should not lead students off campus or directions that may lead them to getting lost around campus. Scholar Path should have the following safeguards:

- Scholar Path should warn about users leaving off campus.
- Notification of any ongoing emergencies on campus such fires, earthquakes, lockdowns as well as safety drills.
- Notification of any safety warning issued by campus police.
- Scholar Path should follow the standards set by Cal State LA campus safety guidelines.

5.3 Security Requirements(CK)

Specify any requirements regarding security or privacy issues surrounding use of the product or protection of the data used or created by the product. Define any user identity authentication requirements. Refer to any external policies or regulations containing security issues that affect the product. Define any security or privacy certifications that must be satisfied.

Scholar Path Database should only allow students with their CIN to access their own schedules and no one else's.

- CIN verification such as providing login information using actual student account information may not be available; however, placeholder test account should be used to ensure that passwords are properly secured and hashed using an algorithm like SHA256.
- Database retrieval should be safe from any data corruption that may occur when a user retrieves data from it.
- Database should be secured from any time of Data breaches that can occur or malicious unauthorized editing of the database.

5.4 Software Quality Attributes (CK)

Scholar Path should be robust and be responsive to students as well as being able to easily be maintained for future semesters:

- Instability such as crashes and freezing should not be a problem with students retrieving information about their classes or browsing through the map.
- Database should be maintainable to allow for updating schedules beyond the fall of 2024 semester.
- The classroom locator should be flexible to allow for changes in case of emergency. maintenance or construction that may cause changes in classroom location or direction.
- Webapp should be portable with the app being able to work on a variety of devices.

5.5 Business Rules

List any operating principles about the product, such as which individuals or roles can perform which functions under specific circumstances. These are not functional requirements in themselves, but they may imply certain functional requirements to enforce the rules.

Scholar path development team should be flexible with their roles in accordance with Agile methods of software engineering.

6. Legal and Ethical Considerations (LG)

Scholar Path will comply with laws and regulations related to data privacy, data protection, and accessibility.

- Compliance with Laws and Regulations -

Scholar Path will make sure to follow the laws and steps taken to ensure compliance by the following:

- Making sure that user data is only accessed with explicit consent.
- Following the terms and conditions of CSULA regarding students' privacy and information.
- Conduct audits on user’s information to make sure nothing is being abused

- Licensing -

Scholar Path will use third-party APIs:

- Leaflet API License: Leaflet is a JavaScript library for interactive maps and can be used freely for personal and commercial purposes. Leaflet terms and conditions must be respected to maintain license.
- Navigator API License: Navigator is included in all modern web browsers. The terms and conditions for the web browsers must be respected to maintain license.

Scholar Path’s own Licensing terms:

- Scholar Path will be licensed under a proprietary software license that restricts unauthorized modification without permission. This will ensure that the software will be used by all students freely but cannot be altered.

- Data Privacy and Security -

Scholar Path will employ authentication to protect user data:

- Authentication: All users must login using their CIN (or Faculty/Staff ID) to access their personal navigation features.

Scholar Path complies with data privacy laws:

- User data will be stored securely and only those with administrative rights will be able to alter/delete user data. Students will still have access to all functionality of the software.

- Ethical Considerations -

User Consent:

- Scholar Path will follow CSULA terms and conditions regarding users information. Guests will have access to a limited version of the software and will have to agree to the terms and conditions.
- The software will notify the users the first time they login and whenever any updates on data collection in the future. All users will be notified what data will be collected and how it will be used.

Transparency:

- Scholar Path will maintain transparency by notifying users on how the software works and being truthful on the applications data collection.
- Scholar Path impact on the user is to simplify campus navigation and provide value to all users.

- Fairness and Non-Discrimination
  - Measures to avoid discrimination

    - Ensuring equal access
- User Safety
    - Risk assessment and mitigation
    - Protection from malicious activities
- Environmental Impact
    - Assessment of environmental impact
    - Steps to minimize footprint

Appendix A: Glossary (KS)

GPS: Global Positioning System

API: Application Programming Interface

UI: User Interface

Map Integration: Combining map data with real-time GPS functionality for navigation

Software: A collection of instructions, data, or computer programs that are used to run machines and carry out activities.

Relational Database: a type of database that stores and provides access to data points that are related to one another.

JavaScript: high-level and interpreted programming language that is widely used for web development.

Object-Oriented Programming: computer programming model that organizes software design around data, or objects, rather than functions and logic.

Logic Design: basic organization of the circuitry of a digital computer

Database: a structured set of data held in a computer, especially one that is accessible in various ways.

Data Security: is the practice of protecting digital information from unauthorized access, corruption or theft throughout its entire lifecycle.

Test Criteria: outlining the software's requirements to be considered successful in testing.

CIN: Campus Identification Number

Appendix B: Analysis Models

Data Flow Diagrams (DFD):

DFD that describes how data moves within Scholar Path eg from input (user requesting routes) to backend (database queries) and output (displaying navigation.)

Class Diagrams:

UML class diagram: This provides a high-level overview of the connections between various classes such as User, MapService, Building and ClassSchedule.

ERD (Entity-Relationship Diagrams) :

For example, an Entity-Relationship Diagram (ERD) is a good starting point showing where different database tables (e.g., Users, Buildings, Rooms and Classes) are referrring to.

State Transition Diagrams:

A state diagram would be of use to clarify some states a user might be if, for example "logged in", "looking up location", "walking there", etc., and the transitions between those states.

Use case diagram- Shows how actors (users) interact with the different usages of the system.

Appendix C: To Be Determined List

GPS Accuracy Settings:

TDB: Define acceptable GPS accuracy based on testing devices from various manufacturers

Map Design:

Outstanding Project: Real-Time Campus Map Visual Final Design

Security Protocol:

TBD: Decide which levels of encryption will be used for transmitting data.

User Feedback System:

Unsure: An in-app real-time user feedback system for users to report navigation inconveniences or system errors.